

CHAPTER – 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 Introduction:

The Myntra Database Management System is designed to manage customers, products, sellers, inventory, orders, payments, reviews, warranties, shipments and administrators efficiently. Functional dependency analysis identifies attribute relationships, while redundancy and anomaly analysis ensure consistency and reduce duplicate data.

1. CUSTOMER TABLE

Functional Dependency:

Customer_ID → First_Name, Last_Name, Gender, Email, Phone, Password, Address, City, State, Pincode, Registration_Date

Redundancy Analysis:

Customer data is stored only once. Other tables reference Customer_ID.

Data Anomalies:

- Insert Anomaly: New customers can be added without orders.
- Update Anomaly: Updating customer details is done only in Customer.
- Delete Anomaly: Deleting an order does not remove customer data.

2. CATEGORY TABLE

Functional Dependency:

Category_ID → Category_Name, Category_Description, Status

Redundancy Analysis:

Categories are stored once.

Data Anomalies:

- Insert Anomaly: New categories can be added.
- Update Anomaly: Rename category once.
- Delete Anomaly: Deleting products doesn't remove category.

3. SELLER TABLE

Functional Dependency:

$\text{Seller_ID} \rightarrow \text{Seller_Name, Contact_Person, Phone, Email, Address, GST_Number}$

Redundancy Analysis:

Seller details stored separately.

Data Anomalies:

- Insert Anomaly: Add seller independently.
- Update Anomaly: Update seller once.
- Delete Anomaly: Deleting product won't delete seller.

4. PRODUCT TABLE

Functional Dependency:

$\text{Product_ID} \rightarrow \text{Product_Name, Category_ID, Seller_ID, Brand, Size, Color, Price, Discount, Stock}$

Redundancy Analysis:

Product details stored once.

Data Anomalies:

- Insert Anomaly: Add product before orders.
- Update Anomaly: Update price once.
- Delete Anomaly: Deleting orders won't delete product.

5. INVENTORY TABLE

Functional Dependency:

Inventory_ID $\rightarrow$ Product_ID, Quantity, Warehouse, Last_Updated

Redundancy Analysis:

Inventory separated from product.

Data Anomalies:

- Insert Anomaly: Create inventory record.
- Update Anomaly: Update stock only here.
- Delete Anomaly: Deleting product record policy aside, inventory independent.

6. ORDERS TABLE

Functional Dependency:

Order_ID $\rightarrow$ Customer_ID, Order_Date, Total_Amount, Order_Status,
Shipping_Address, Delivery_Date

Redundancy Analysis:

Order info separated.

Data Anomalies:

- Insert Anomaly: Create order for customer.
- Update Anomaly: Update status once.
- Delete Anomaly: Deleting order won't remove customer.

7. ORDER ITEMS TABLE

Functional Dependency:

Order_Item_ID $\rightarrow$ Order_ID, Product_ID, Quantity, Unit_Price, Discount, Total_Price

Redundancy Analysis:

Avoids repeating products in orders.

Data Anomalies:

- Insert Anomaly: Add items independently.
- Update Anomaly: Update quantity here.
- Delete Anomaly: Deleting item won't delete product.

8. WARRANTY TABLE**Functional Dependency:**

Warranty_ID → Product_ID, Warranty_Period, Warranty_Start, Warranty_End,
Warranty_Status

Redundancy Analysis:

Warranty stored separately.

Data Anomalies:

- Insert Anomaly: Warranty added after purchase.
- Update Anomaly: Update warranty once.
- Delete Anomaly: Deleting warranty doesn't delete product.

9. PAYMENT TABLE**Functional Dependency:**

Payment_ID → Order_ID, Payment_Method, Transaction_ID, Payment_Status,
Payment_Date, Amount

Redundancy Analysis:

Payment stored separately.

Data Anomalies:

- Insert Anomaly: Payment after order.

- Update Anomaly: Update payment status once.
- Delete Anomaly: Deleting payment doesn't delete order.

10. REVIEW TABLE

Functional Dependency:

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis:

Reviews separate.

Data Anomalies:

- Insert Anomaly: Review after purchase.
- Update Anomaly: Edit review only here.
- Delete Anomaly: Deleting review doesn't remove product.

11. SHIPMENT TABLE

Functional Dependency:

Shipment_ID → Order_ID, Courier_Name, Tracking_Number, Shipment_Status, Dispatch_Date, Delivery_Date

Redundancy Analysis:

Shipment stored separately.

Data Anomalies:

- Insert Anomaly: Shipment after order.
- Update Anomaly: Update tracking once.
- Delete Anomaly: Deleting shipment doesn't remove order.

12. ADMIN TABLE

Functional Dependency:

Admin_ID → Admin_Name, Email, Phone, Role, Username, Password

Redundancy Analysis:

Admin data separate.

Data Anomalies:

- Insert Anomaly: Add admin independently.
- Update Anomaly: Update admin once.
- Delete Anomaly: Deleting admin doesn't affect business data.

Conclusion

The Myntra Database Management System follows proper functional dependencies and normalization principles. Each entity stores only relevant data, reducing redundancy and preventing insert, update and delete anomalies. This improves data integrity, consistency and efficient database management.